

## ENTER RANDOMNESS

Things can become more interesting when randomness enters the game. Imagine a waiting room full of actors queuing for an audition. The number of actors who will win is clearly small, and they are the ones generally observed by the public as representative of the profession, as we saw in our discussion on survivorship bias. The winners would move into Bel Air, feel pressure to acquire some basic training in the consumption of luxury goods and, perhaps owing to the dissolute and unrhythmic lifestyle, flirt with substance abuse. As to the others (the great majority), we can imagine their fate; a lifetime of serving foamed *caffè latte* at the neighboring Starbucks, fighting the biological clock between auditions.

One may argue that the actor who lands the lead role that catapulted him into fame and expensive swimming pools has some skills others lack, some charm, or a specific physical trait that was a perfect match for such a career path. I beg to differ. The winner may have had some acting skills, but so do all of the others, otherwise they would not have been in the waiting room.

It is an interesting attribute of fame that it has its own dynamics. An actor becomes known by some parts of the public because he is known by other parts of the public. The dynamics of such fame follow a rotating helix, which may have started at the audition, as the selection could have been caused by some silly detail that fitted the mood of the examiner on that day. Had the examiner not fallen in love the previous day with a person of similar sounding last name, then our selected actor from that particular sample *history* would be serving *caffè latte* in the intervening sample *history*.

## LEARNING TO TYPE

Researchers frequently use the example of QWERTY to describe the vicious dynamics of winning and losing in an economy, and to illustrate how the final outcome is more than frequently the undeserved one. The arrangement of the letters on a typewriter is an example of the success of the least deserving method. For our typewriters have the orders of the

letters on their keyboard arranged in a non-optimal manner, as a matter of fact in such a non-optimal manner as to slow down the typing rather than make the job easy, in order to avoid jamming the ribbons as they were designed for less electronic days. Therefore, as we started building better typewriters and computerized word processors, several attempts were made to rationalize the computer keyboard, to no avail. People were trained on a QWERTY keyboard and their habits were too sticky for change. Just like the helical propulsion of an actor into stardom, people patronize what other people like to do. Forcing a rational dynamic on the process would be superfluous, nay, impossible. This is called a *path dependent outcome*, and has thwarted many mathematical attempts at modeling behavior.

It is obvious that the information age, by homogenizing our tastes, is causing the unfairness to be even more acute – those who win capture almost all the customers. The example that strikes most as the most spectacular lucky success is that of the software maker Microsoft and its moody founder Bill Gates. While it is hard to deny that Gates is a man of high personal standards, work ethics, and above average intelligence, is he the best? Does he *deserve* it? Clearly not. Most people are equipped with his software (like myself) because other people are equipped with his software, a purely circular effect (economists call that “network externalities”). Nobody ever claimed that it was the best software product. Most of Gates’s rivals have an obsessive jealousy of his success. There are maddened by the fact that he managed to win so big while many of them are struggling to make their companies survive.

Such ideas go against classical economic models, in which results either come from a precise reason (there is no account for uncertainty) or the good guy wins (the good guy is the one who is more skilled and has some technical superiority). Economists discovered path-dependent effects late in their game, then tried to publish wholesale on the topic that otherwise would be bland and obvious. For instance, Brian Arthur, an economist concerned with nonlinearities at the Santa Fe Institute, wrote that chance events coupled with positive feedback rather than technological superiority will determine economic superiority – not some abstrusely defined edge in a given area of expertise. While early economic models excluded randomness, Arthur explained how “unexpected orders, chance meetings with lawyers, managerial whims